AI-Powered Security Operations Center (SOC) Assistant Using Large Language Models

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PPD-2 — Second Review Project Representation

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Problem Statement & Objectives

Operational Bottlenecks in Modern Security Operations Centers

SOC Operational Bottlenecks

- **Alert Inundation**: Tier-1 analysts handle thousands of raw events per shift; repetitive triage causes analyst fatigue and critical breach misses.
- **Manual Attribution Overhead**: Manually correlating indicators of compromise (IoCs) across MITRE ATT&CK and NIST NVD slows Mean Time to Detect (MTTD).
- **Hallucination in Cloud LLMs**: Generic models fabricate non-existent CVEs and breach corporate data privacy through external cloud API calls.

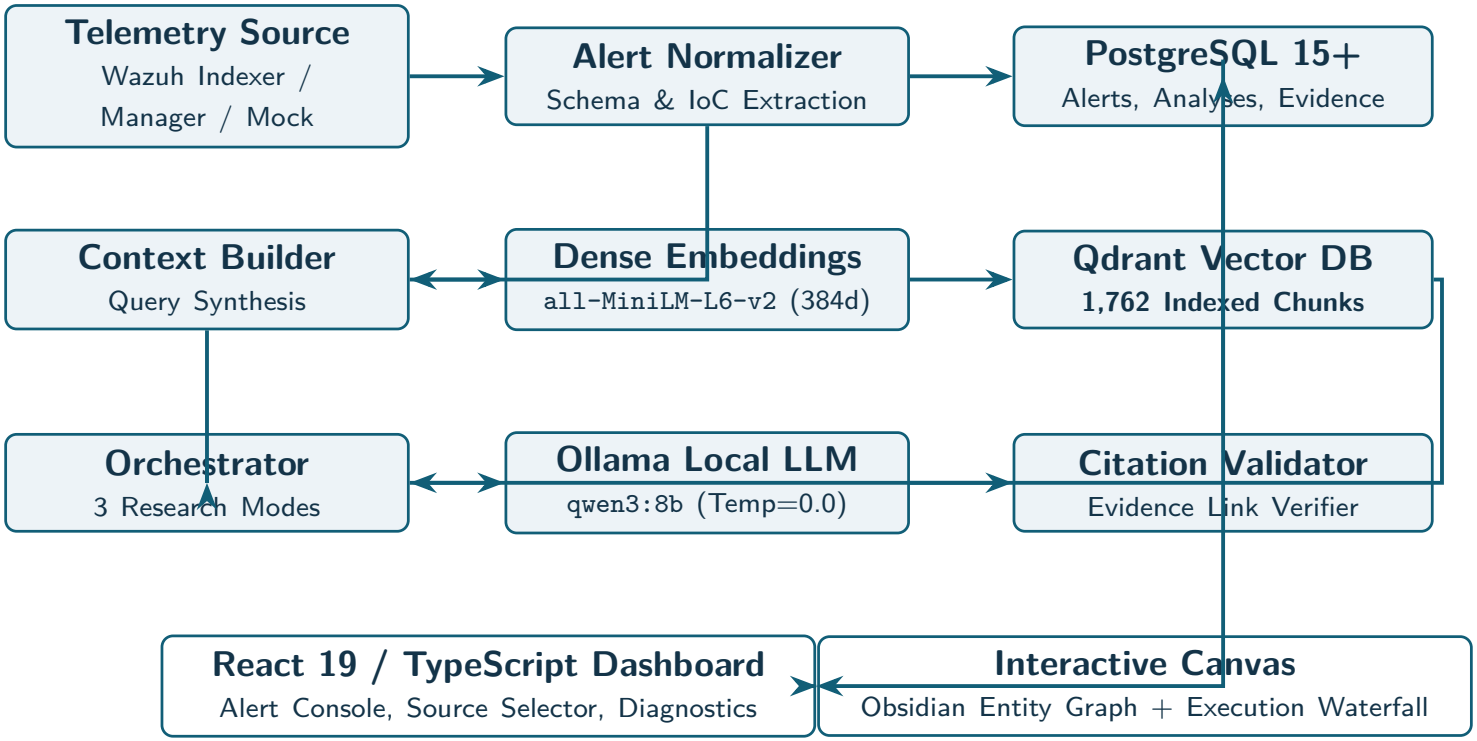
Core Research Objectives

- **Evidence-Grounded RAG**: Inject structured threat intelligence (1,762 vector chunks) into local LLM reasoning to anchor advisory outputs.
- **On-Premise Privacy**: Run entirely local using Ollama (`qwen3:8b`) and Qdrant without sending telemetry to external clouds.
- **Auditable Provenance**: Enforce programmatic validation where every claim must cite verified retrieved evidence IDs.
- **Human-in-the-Loop**: Produce structured triage recommendations rather than unverified automated remediation.

Scope Boundary: The system functions as an evidence-backed Tier-1 advisory copilot; final remediation authorization remains strictly with human SOC analysts.

1. System/Design Development

Multi-Tier Security Operations Assistant Architecture



Ingestion: Multi-source isolation stamps demo vs. live telemetry.

Knowledge: 1,618 MITRE ATT&CK + 144 NVD CVE vector chunks.

Privacy: 100% on-premise execution with zero external cloud leakage.

6-Stage Sequential Operational Pipeline

1. **Ingestion & Normalization**: Parses Wazuh JSON/Syslog into normalized schema; extracts timestamps, agent, rule ID, and network IoCs.
2. **Context Extraction & Query Synthesis**: Synthesizes 3 targeted queries: rule text query, MITRE technique anchor, and entity context.
3. **Dense Semantic Search (Qdrant)**: Computes 384-d cosine similarity over 1,762 chunks; retrieves top-k evidence cards.
4. **Evidence-Constrained Prompting**: Formats prompt with strict system instructions prohibiting uncited assertions.
5. **Deterministic LLM Reasoning**: Local `qwen3:8b` evaluates threat scenario and formulates triage recommendations.
6. **Citation Integrity Verification**: Automated validator verifies cited evidence IDs match retrieved items before database write.

Stage Latency Instrumentation

Measured Pipeline Durations:

- Ingestion & Parse: 1–5 ms
- Normalization & DB: 2–8 ms
- MiniLM Embeddings: 30–60 ms
- Qdrant Vector Search: 8–25 ms
- Ollama Inference (`qwen3:8b`): 35–80 s
- Citation Validation: 1–3 ms

Integrity Guarantee:

Uncited claims are flagged, ensuring transparent auditability for human analysts.

2. Implementation Progress

Subsystem Development Status & Infrastructure Overview

Subsystem	Component Technology	Implementation Details & Verification Evidence	Status
Backend Core	FastAPI (Python 3.11/13)	Asynchronous REST API, typed Pydantic V2 schemas, SSE trace streaming	**Verified**
Relational DB	PostgreSQL 15+ / Alembic	Tables: `alerts` (95 records), `analyses` (44 records), `evidence` (48 records)	**Verified**
Vector DB	Qdrant Engine (v1.18+)	Collection `security_knowledge` active with **1,762 indexed vector points**	**Verified**
Embeddings	`all-MiniLM-L6-v2`	384-dimensional dense semantic vectors, local PyTorch inference	**Verified**
Local LLM	Ollama / `qwen3:8b`	8B parameter local reasoning at temperature 0.0; JSON schema enforcement	**Verified**
Frontend UI	React 19 + TypeScript	Single-page app with Obsidian Entity Graph and Execution Waterfall traces	**Verified**
Source Manager	Isolated 3-Mode Abstraction	Demo (active default), Live Wazuh (clean empty state), Dataset Replay	**Verified**
Wazuh Transport	Indexer & Manager Clients	Read-only OpenSearch DSL (`POST /wazuh-alerts-*/_search`) & JWT manager	**Client Verified**
Deployment	Docker Compose	Multi-service container orchestration with internal health probes	**Verified**

Live Service Probe Verified (2026-10-26 13:16 IST): `GET /health` → `dependencies_ok: true`, **PostgreSQL: connected (68ms), Qdrant: connected (1,762 points), Ollama: reachable (`qwen3:8b`).**

Operational Analyst Modules

- **SOC Executive Dashboard**: Real-time KPI summary, severity donut distribution (25 Critical, 52 High, 12 Medium, 6 Low), and component health matrix.
- **Alert Management Console**: Tabular alert stream with severity badges, pagination, status updates, and deep inspector links.
- **AI Investigation Workspace**: Multi-mode analysis triggering, structured threat synthesis, recommendations, and confidence indicator.
- **Data Source Selector**: Top command-bar segmented selector switching between Demo Data, Live Wazuh, and Replay.

Advanced Analytical Workspaces

- **Obsidian-Style Entity Graph**: Force-directed interactive canvas linking alerts, hostnames, IP addresses, usernames, and MITRE techniques (`EntityGraphPage.tsx`).
- **Pipeline Execution Waterfall**: Visual timeline inspector tracking per-stage latencies and intermediate outputs (`ExecutionTracesPage.tsx`).
- **Evaluation Benchmark UI**: Progress tracker displaying real-time metrics across test runs (`Evaluation.tsx`).
- **Honest Empty States**: Dedicated UI disclosure when Live Wazuh is selected without configured endpoints.

UI Verification Status: 32 passed frontend unit tests across 8 suites (Vitest v5.0.1); production build compiles cleanly in 4.91s.

Backend Test Suite (Pytest v9.1.1)

- **Total Passed**: **226 Passed**, 0 Failed (44.51s)
- **Environment**: Python 3.11/3.13 in Docker container
- **Verified Functional Domains**:
  - Ingestion, deduplication & persistence (5 tests)
  - Data source isolation & Indexer client (5 tests)
  - Knowledge chunking & embeddings (53 tests)
  - Vector store indexing & search (8 tests)
  - Security context & query extraction (28 tests)
  - Orchestrator modes & validation (27 tests)
  - Evaluation metrics & resume runner (33 tests)
  - Wazuh transport & HTTP client (15 tests)

Frontend Test Suite (Vitest v5.0.1)

- **Total Passed**: **32 Passed**, 0 Failed (27.01s)
- **Environment**: React 19, TypeScript, Vite
- **Verified Component Suites**:
  - `Dashboard`: Metric rendering & empty states
  - `Alerts`: Filtering, search & provenance chips
  - `AIAnalysis`: Mode selection & error recovery
  - `EntityGraph`: Canvas interaction & nodes
  - `ExecutionTraces`: Waterfall & drawer
  - `Evaluation`: Benchmark run tracking
  - `Layout`: Navigation & source selector
  - `App`: Router configuration & fallbacks

Knowledge Base Construction

- **MITRE ATT&CK Enterprise (v14)**: Extracted 1,618 techniques, tactics, mitigation steps, and detection guidelines.
- **NIST NVD CVE Security Feed**: Extracted 144 high-impact CVE entries with CVSS scores, affected software, and remediation paths.
- **Chunking & Indexing**: 1,200 character chunks with 150 character overlap; indexed into Qdrant using 384-d cosine similarity.
- **Deterministic Embeddings**: Local `all-MiniLM-L6-v2` model running on CPU/GPU without cloud calls.

Multi-Query Retrieval Strategy

- **Rule Query**: Synthesizes query directly from alert rule description to identify relevant attack categories.
- **Technique Anchor**: Injects explicit MITRE technique tags from security context to anchor top-1 retrieval.
- **Entity Context**: Leverages host, user, and network indicators for targeted vulnerability correlation.
- **Strict Citation Validation**: Rejects or flags assertions lacking explicit evidence IDs.

Three Strictly Isolated Data Sources

- **Mode 1 — Demo (`demo`)**: 95 persistent records (52 attack scenario alerts, 40 benchmark cases). Serves as dashboard and evaluation baseline.
- **Mode 2 — Live Wazuh (`live_wazuh`)**: Real alerts from configured Wazuh Indexer or Manager. Displays 0 alerts and honest empty state until connected.
- **Mode 3 — Dataset Replay (`dataset_replay`)**: Offline benchmark datasets (e.g. CIC-IDS2017); explicitly labeled as historical replay.

Wazuh Connector Design

- **Preferred: Wazuh Indexer API**: Queries OpenSearch cluster directly (`POST /wazuh-alerts-*/_search`) over TLS with HTTP Basic Auth.
- **Wazuh Manager API**: REST API client with automatic JWT token acquisition and refresh (`GET /alerts`).
- **Read-Only Guarantees**: Connectors perform zero write, delete, or re-indexing operations on remote Wazuh deployments.
- **Credential Isolation**: All secrets stored in `.env`; never committed or exposed in UI bundles.

Controlled 3-Mode Experiment Design

- **Mode 1 (`llm_only`)**: Baseline. Alert passed to `qwen3:8b` without external retrieval; measures raw model memory.
- **Mode 2 (`standard_rag`)**: Unanchored semantic search using rule description text; tests basic vector retrieval.
- **Mode 3 (`evidence_aware`)**: Multi-query synthesis, entity anchoring, top-k evidence ranking, and strict citation verification.

Evaluation Datasets & Scenarios

- **Dataset v1 (Smoke Baseline)**: 8 multi-stage scenarios across brute-force, web attacks, and privilege escalation.
- **Dataset v2 (Benchmark)**: 40 curated scenarios across 7 attack categories + benign controls ($40 \times 3 = 120$ planned evaluations).
- **Benign Control Scenarios**: Evaluates false alarm resistance on routine administrative events.

Evaluated Metric Suite: Technique Precision/Recall/F1, CVE Precision/Recall/F1, Valid Citation Rate, Retrieval Recall@1, Recall@5, MRR, Wall Latency, and LLM Inference Latency.

4. Preliminary Results
Empirical Findings from Controlled Benchmark Runs

Benchmark Execution Status (Dataset v2, Run `20261024-145621-1345d6`): 40 scenarios $\times$ 3 modes = 120 planned evaluations. Currently 12 scenarios completed (36 evaluations completed), with 84 evaluations pending execution.

Evaluation Mode	Cases (N)	Avg Wall Latency	Technique Prec.	Technique Recall	Technique F1	Valid Citation Rate
LLM-Only (Baseline)	12	36.8 s	0.500	0.417	0.444	N/A (no citations)
Standard RAG	12	82.2 s	0.500	0.417	0.444	**1.000 (100%)**
Evidence-Aware RAG	12	79.6 s	0.500	0.417	0.444	**1.000 (100%)**

Dataset v1 Baseline ($N = 8$, Completed)

- **Retrieval Recall@1**: **1.000** (Evidence-Aware) vs **0.250** (Standard RAG) vs 0.000 (LLM-Only).
- **Retrieval MRR**: **1.000** vs **0.354** for Standard RAG.
- **Technique F1**: 0.833 across 8 smoke scenarios.

Retrieval-Only Ablation (No LLM)

- **Semantic Anchored**: Recall@1 = **1.000**, MRR = **1.000** (1563 ms).
- **Semantic Unanchored**: Recall@1 = 0.375, MRR = 0.463 (338 ms).
- **Key Takeaway**: Entity anchoring dramatically improves top-1 retrieval accuracy.

Limitations & Challenges

Transparent Analysis of Current Engineering Constraints

Identified Technical Limitations

- **Simulated Evaluation Scenarios**: Benchmarks currently utilize curated synthetic scenarios; validation on noisy enterprise production logs remains ongoing.
- **Inference Latency on Local Hardware**: Local `qwen3:8b` inference requires 35–80s per alert on host hardware; presents challenges for real-time high-throughput triage.
- **Pending Live Wazuh Deployment**: Wazuh Indexer and Manager client code is fully verified, but evaluation currently operates in mock

Integrity & Engineering Mitigations

- **Checkpoint Resume Protocol**: Evaluator saves intermediate state to `results_partial.json`, preventing data loss on interrupted runs.
- **Explicit UI Disclaimers**: Prominent badge disclosures inform analysts of mock/synthetic data mode.
- **Exact vs. Semantic Scoring**: Current scorer requires exact MITRE ID matches; related parent techniques receive zero credit, representing conservative lower-bound metrics.

Future Work & Next Milestones

Planned Research Activities for Final Review

01 COMPLETE 120-EVALUATION BENCHMARK

Execute remaining 84 scenario/mode evaluations on Dataset v2 to establish aggregate statistical significance.

02 REMOTE LIVE WAZUH SIEM CONNECTION

Deploy remote Wazuh server and connect live telemetry via `.env.wazuh.example` configuration template.

03 MODEL QUANTIZATION & DISTILLATION

Evaluate quantized 3B/4B models (e.g. Qwen-2.5-3B-Instruct) to reduce local inference latency below 10 seconds.

04 ANALYST FEEDBACK & REINFORCEMENT LOOP

Implement interactive thumbs up/down analyst feedback capture to refine retrieval ranking and prompt templates.

Summary of Review Deliverables

- **System/Design Development**: Architected a fully containerized, multi-tier SOC Assistant integrating PostgreSQL, Qdrant, and Ollama.
- **Technical Approach**: Engineered an evidence-aware RAG pipeline indexing 1,762 MITRE ATT&CK and NVD chunks with strict citation validation.
- **Implementation Progress**: Developed React 19 UI with Entity Graph and Waterfall traces, verified by **226 backend and 32 frontend unit tests**.
- **Preliminary Results**: Established 100% valid citation integrity and demonstrated superior top-1 retrieval precision (Recall@1 = 1.0 vs 0.25).

Thank You!

Questions & Discussion

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Status: PPD-II Second Review Completed

1. Ismail et al., "Enhancing Security Operations Center: Wazuh Security Event Response with Retrieval-Augmented-Generation-Driven Copilot", *Sensors*, 2025.

2. Habibzadeh, Feyzi & Ebrahimi Atani, "Large Language Models for Security Operations Centers: A Comprehensive Survey", *Journal of Electrical and Computer Engineering*, 2026.

3. Taysi et al., "Evidence-First Retrieval Augmented Generation with Claim Repair for Security Operations Center Workflows", *IEEE ICHORA*, 2026.

4. Hahn et al., "Non-Disruptive Disruption: An Empirical Experience of Introducing LLMs in the SOC", *NDSS / Internet Society*, 2026.

5. MITRE Corporation, "MITRE ATT&CK Enterprise Framework (Version 14)", `https://attack.mitre.org/`, 2023.

6. National Institute of Standards and Technology (NIST), "National Vulnerability Database (NVD) Data Feeds", `https://nvd.nist.gov/`, 2024.